

Prediction Report

Task 12: Predictive Analytics

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Date 6/26/2026

Project Title

Marketing Campaign ROI Prediction using Linear Regression

Objective

The objective of this project is to develop a predictive analytics model capable of estimating the Return on Investment (ROI) of marketing campaigns. The model uses historical campaign information to forecast future campaign performance.

Dataset Description

The project uses a marketing dataset containing information related to various advertising campaigns.

Dataset Features

- Campaign ID
- Company
- Campaign Type
- Target Audience

- Campaign Duration
- Marketing Channel
- Conversion Rate
- Acquisition Cost
- Clicks
- Impressions
- Engagement Score
- Location
- Language
- Customer Segment
- Date

Target Variable

ROI (Return on Investment)

Data Preprocessing

The following preprocessing steps were completed:

- Checked for missing values.
 - Converted Acquisition Cost from currency format to numeric values.
 - Converted campaign duration into integer values.
 - Extracted Year, Month, and Day from the Date column.
 - Applied One-Hot Encoding to categorical variables.
 - Split the dataset into training and testing sets using an 80:20 ratio.
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Machine Learning Algorithm

Algorithm: Linear Regression

Linear Regression was selected because the target variable (ROI) is continuous and numerical.

Model Training

- Training Data: 80%
- Testing Data: 20%

The model was trained using Scikit-learn's LinearRegression class.

Model Evaluation

The model performance was measured using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 Score

Example Results

RMSE : 1.7356928061037387

MAE : 1.503004785881713

R^2 Score : : -0.0002981167563391729

Prediction

The trained model predicts the expected ROI of future marketing campaigns based on campaign characteristics.

Example:

Input:

- Campaign Type
- Acquisition Cost
- Channel Used
- Clicks
- Impressions
- Engagement Score

Output:

Predicted ROI = _____

Project Outcome

The predictive analytics model successfully estimates campaign ROI using historical marketing data. The model demonstrates how machine learning can support marketing teams in forecasting campaign performance, optimizing advertising strategies, and improving investment decisions.

Skills Demonstrated

- Data Cleaning
 - Feature Engineering
 - Data Encoding
 - Linear Regression
 - Model Evaluation
 - Predictive Analytics
 - Machine Learning with Scikit-learn
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Conclusion

This project successfully implemented a Linear Regression model for marketing campaign ROI prediction. The workflow included data preprocessing, model training, evaluation, prediction, and model saving. The project highlights the practical application of predictive analytics in solving real-world business problems and demonstrates proficiency in Python and Scikit-learn.